

PROVISIONAL PATENT APPLICATION

SYNTHEGY

Multi-Database Knowledge Graph Traversal Method for Chemical Compound-to-Disease Reasoning

A method and system for automatically traversing multiple heterogeneous chemical and biological databases in a single API call to construct a knowledge graph linking a chemical compound to its biological targets, metabolic pathways, and associated diseases, with association scoring and natural-language reasoning over the resulting graph.

Inventor: [To be completed by filer]

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TITLE OF INVENTION

Multi-Database Knowledge Graph Traversal Method for Automated Chemical
Compound-to-Disease Biological Context Reasoning

CROSS-REFERENCE TO RELATED APPLICATIONS

Not applicable.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to computational chemistry and bioinformatics, and more specifically to a method for automatically traversing multiple heterogeneous databases to construct a knowledge graph that links a chemical compound to its biological targets, metabolic pathways, and associated diseases, enabling LLM-based reasoning over the biological context of a compound in a single API call.

Description of Related Art

Chemistry-AI platforms such as Schrodinger, Exscientia, and Recursion Pharmaceuticals provide computational tools for drug discovery, but none automatically connect a chemical compound to its disease associations through a multi-database traversal. Existing approaches require manual lookup across separate databases (PubChem for compound properties, ChEMBL for bioactivity, KEGG for pathways, OpenTargets for disease associations), each with different query languages (REST, GraphQL) and different identifier systems (PubChem CID, ChEMBL ID, gene symbols, Ensembl IDs, MONDO disease IDs).

The problem is one of identifier resolution and cross-database traversal: a PubChem CID (e.g. 2244 for aspirin) must be mapped to a ChEMBL ID (ChEMBL25), which must be mapped to a gene symbol (PTGS2), which must be mapped to an Ensembl ID (ENSG00000073756), which must be queried against OpenTargets to retrieve disease associations with scores. No existing system performs this full traversal automatically.

What is needed is a method that, given a chemical compound identifier, automatically constructs a knowledge graph spanning compound, target, pathway, and disease databases, with association scoring and natural-language reasoning capability over the resulting graph.

SUMMARY OF THE INVENTION

The present invention provides a computer-implemented method for constructing a knowledge graph that links a chemical compound to its biological context, comprising: (a) receiving a chemical compound identifier; (b) querying a first database to retrieve compound properties; (c) querying a second database to identify biological targets of the compound; (d) extracting gene symbols from target names using a predefined alias mapping; (e) querying a third database to identify metabolic pathways associated with the compound; (f) querying a fourth database to identify diseases associated with the targets, with association scores; (g) constructing a graph data structure with nodes representing compounds, targets, pathways, and diseases, and edges representing the relationships between them; and (h) returning the graph to a client for natural-language reasoning.

The method further includes caching database responses with time-to-live values appropriate to the data's mutability, and an active-learning feedback loop that records user accept/revise signals on reasoning outputs and extracts keyword patterns to create a proprietary preference profile.

DETAILED DESCRIPTION

System Architecture

The system comprises a frontend application, an API gateway with authentication and rate limiting, and a plurality of microservices, each wrapping a different public database. The microservices communicate via HTTP REST and GraphQL protocols. A Python subprocess bridge invokes RDKit for molecular descriptor computation and ord_schema for Open Reaction Database parsing.

The Traversal Method

The core inventive step is the multi-database traversal method, which proceeds as follows:

Step 1: Compound Lookup

Given a compound name or SMILES string, the system queries PubChem PUG REST to retrieve canonical SMILES, molecular formula, molecular weight, InChIKey, XLogP, TPSA, and other computed properties. The response is cached in SQLite with a 30-day TTL.

Step 2: Target Identification

The system queries ChEMBL REST API to identify biological targets associated with the compound. For each target, the system extracts a gene symbol using a predefined alias mapping table (e.g. 'Cyclooxygenase' maps to 'PTGS2', 'COX-2' maps to 'PTGS2', 'Epidermal Growth Factor Receptor' maps to 'EGFR'). The alias table covers common protein name variants, enzymatic names, and receptor subclass names.

Step 3: Ensembl ID Resolution

For each extracted gene symbol, the system queries the OpenTargets GraphQL API to resolve the corresponding Ensembl gene ID (e.g. PTGS2 maps to ENSG00000073756). This step bridges the ChEMBL identifier space to the OpenTargets identifier space.

Step 4: Pathway Linking

The system queries the KEGG REST API to find metabolic pathways associated with the compound. First, the compound name is searched against the KEGG compound database to obtain a KEGG compound ID. Then, the KEGG link endpoint is queried to retrieve all pathway IDs associated with that compound. Pathway names are resolved via the KEGG list endpoint.

Step 5: Disease Association

For each target with a resolved Ensembl ID, the system queries the OpenTargets GraphQL API to retrieve associated diseases with association scores (0.0-1.0). The

results are deduplicated by disease ID and sorted by score. The top N diseases are retained.

Step 6: Graph Construction

The system constructs a graph data structure with four node types (compound, target, pathway, disease) and three edge types ('active against', 'involved in', 'treats/associates'). The graph is returned as a JSON object with nodes, edges, and flat lists for frontend rendering.

Step 7: LLM Reasoning

The graph, along with compound properties, ADMET descriptors, and experimental reaction data, is injected into an LLM prompt. The LLM uses the biological context to produce strategy-aware reasoning about synthesis routes, including which diseases the compound is relevant to and whether the route is appropriate for the therapeutic context.

Active Learning Feedback Loop

The system records user feedback signals (accept, revise, reject) on each LLM reasoning output. Feedback is stored in a database with the original instruction text. The system extracts keyword frequency from accepted versus revised instructions to identify preference patterns (e.g. 'chemist prefers instructions mentioning: protecting, groups, convergent' vs 'chemist tends to revise when instructions mention: toxic, reagents'). The preference profile is used to bias future LLM prompts toward historically preferred patterns.

CLAIMS

Claim 1. A computer-implemented method for constructing a knowledge graph linking a chemical compound to biological context, comprising: receiving a chemical compound identifier; querying a first database to retrieve compound properties; querying a second database to identify one or more biological targets of said compound; extracting one or more gene symbols from said biological targets using a predefined alias mapping; querying a third database to identify one or more metabolic pathways associated with said compound; querying a fourth database to identify one or more diseases associated with said one or more biological targets, each disease having an association score; constructing a graph data structure comprising nodes representing said compound, said biological targets, said metabolic pathways, and said diseases, and edges representing relationships therebetween; and returning said graph data structure.

Claim 2. The method of claim 1, wherein said extracting gene symbols comprises looking up said biological target names in a predefined alias mapping table that maps protein common names, enzymatic names, and receptor subclass names to standard gene symbols.

Claim 3. The method of claim 1, further comprising resolving said gene symbols to Ensembl gene identifiers by querying a fifth database (OpenTargets) with said gene symbols, and wherein said querying a fourth database to identify diseases uses said Ensembl gene identifiers.

Claim 4. The method of claim 1, wherein said first database is PubChem, said second database is ChEMBL, said third database is KEGG, and said fourth database is OpenTargets, and wherein said querying uses REST and GraphQL protocols respectively.

Claim 5. The method of claim 1, further comprising caching database responses in a local cache with time-to-live values, wherein compound properties are cached for 30 days, pathway data for 30 days, and disease associations for 7 days.

Claim 6. The method of claim 1, further comprising providing said graph data structure to a large language model (LLM) as context for natural-language reasoning about chemical synthesis strategies, wherein said LLM produces a strategy score and explanatory text referencing said biological context.

Claim 7. The method of claim 6, further comprising recording user feedback signals on said LLM output, extracting keyword patterns from accepted versus

rejected outputs, and biasing future LLM prompts based on said keyword patterns.

Claim 8. The method of claim 1, wherein said graph data structure is rendered as an interactive visualisation with color-coded node types and association-score-weighted edges, and wherein said diseases are displayed with said association scores as progress bars.

Claim 9. A system comprising a frontend application, an API gateway, and a plurality of microservices, each microservice wrapping a different public database, wherein said system performs the method of claim 1.

Claim 10. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform the method of claim 1.

ABSTRACT

A method and system for automatically traversing multiple heterogeneous chemical and biological databases to construct a knowledge graph linking a chemical compound to its biological targets, metabolic pathways, and associated diseases. The method receives a compound identifier, queries PubChem for properties, ChEMBL for targets, extracts gene symbols via alias mapping, queries KEGG for pathways, and OpenTargets for disease associations with scores. The resulting graph is provided to an LLM for strategy-aware reasoning. An active-learning feedback loop records user preferences and biases future reasoning outputs.

FILING INSTRUCTIONS

1. Go to <https://patents.gov.uk> (UK) or <https://www.uspto.gov> (US)
2. File as a Provisional Patent Application (not a full utility patent)
3. Upload this document as the specification
4. Pay the filing fee (UK: GBP 60, US: \$65 micro entity)
5. You get 'Patent Pending' status immediately
6. You have 12 months to file the full (non-provisional) application
7. Recommended: have a patent attorney review before filing the full application